

Equidistant Point on an Axis

Core1 independently reconstructs the mathematics: object, dependencies, equation anatomy, representations, validity and misconception boundaries.

This sample demonstrates:

- The object is axis constraint + equal-distance locus, not a midpoint recipe.
- Equation authority comes from $PA=PB \rightarrow PA^2=PB^2$.
- The symmetry shortcut is a derived special case with a visible validity condition.
- Core1 hands semantic claims downstream; it does not choose final learner pages.

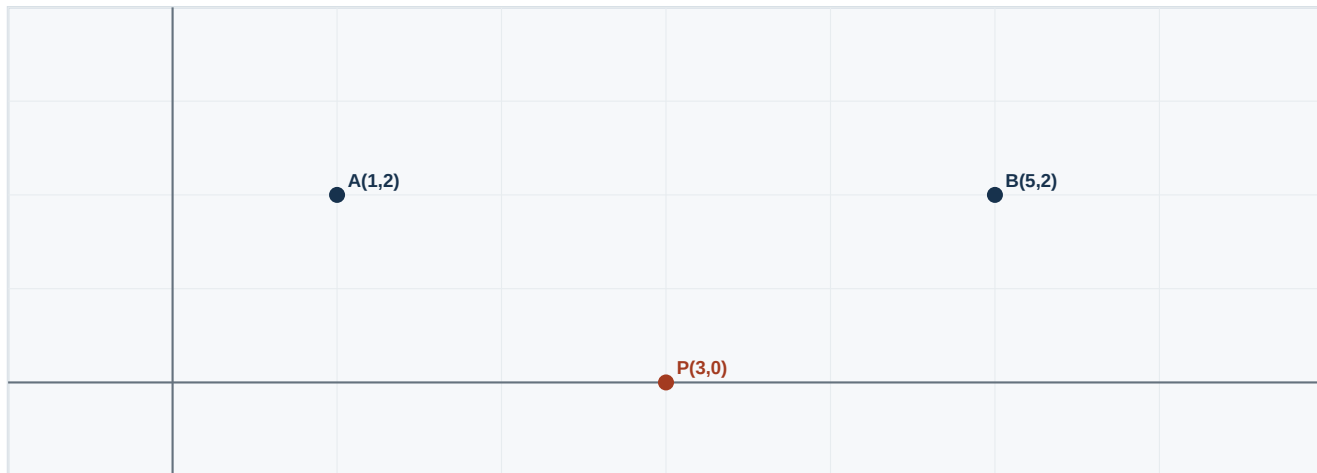
Sample topic

A point on an axis equidistant from two fixed points

Generated review fixture - not an official exam source

1. Object, invariant and representation

What is mathematically true before pedagogy is designed?

**K-CONCEPT**

P is constrained to one axis; one coordinate is fixed.

K-INVARIANT

$PA=PB$, equivalently $PA^2=PB^2$ because distances are non-negative.

K-REPRESENTATION

x-axis $\rightarrow P=(x,0)$; y-axis $\rightarrow P=(0,y)$.

2. Dependency + equation anatomy

No important equation is allowed to appear as a naked rule.

PREREQUISITE

Ordered-pair and axis semantics.

PARENT LAW

$$d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2.$$

MODEL

$$(x - a)^2 + b^2 = (x - c)^2 + d^2.$$

WHY LEGITIMATE

Squared distances preserve equality for non-negative distances.

INTERPRETATION

Solution is the axis point on the perpendicular-bisector locus.

CHECK

Substitute the final point and verify $PA^2 = PB^2$.

3. Validity + misconception boundaries

Shortcuts are admissible only when their structural condition is named.

SPECIAL CASE

$A=(a,b)$, $B=(c,b)$, $P=(x,0)$ $\rightarrow$ common b^2 cancels $\rightarrow x=(a+c)/2$.

VALIDITY LIMIT

The x-midpoint shortcut is not general when A and B have different heights.

WRONG CHAIN

"P is on x-axis, so average x-coordinates."

CORRECT CHAIN

Represent P $\rightarrow$ compare squared distances $\rightarrow$ simplify $\rightarrow$ solve $\rightarrow$ check.

REPRESENTATION RISK

A symmetry sketch may over-suggest midpoint use unless unequal-height contrast is also shown.